presence of a dark-robed mime even though it sees only his white-gloved hands flitting to and fro on the unlit stage, astronomers concluded that the universe must be suffused with *dark matter*—matter that does not clump together in stars and hence does not give off light, and that thus exerts a gravitational pull without revealing itself visibly. The universe's luminous constituents—stars— were revealed as but floating beacons in a giant ocean of dark matter.

But if dark matter must exist in order to produce the observed motions of stars and galaxies, what's it made of? So far, no one knows. The identity of the dark matter remains a major, looming mystery, although astronomers and physicists have suggested numerous possible constituents ranging from various kinds of exotic particles to a cosmic bath of miniature black holes. But even without determining its composition, by closely analyzing its gravitational effects astronomers have been able to determine with significant precision how much dark matter is spread throughout the universe. And the answer they've found amounts to about 25 percent of the critical density.[21] Thus, together with the 5 percent found in visible matter, the dark matter brings our tally up to 30 percent of the amount predicted by inflationary cosmology.

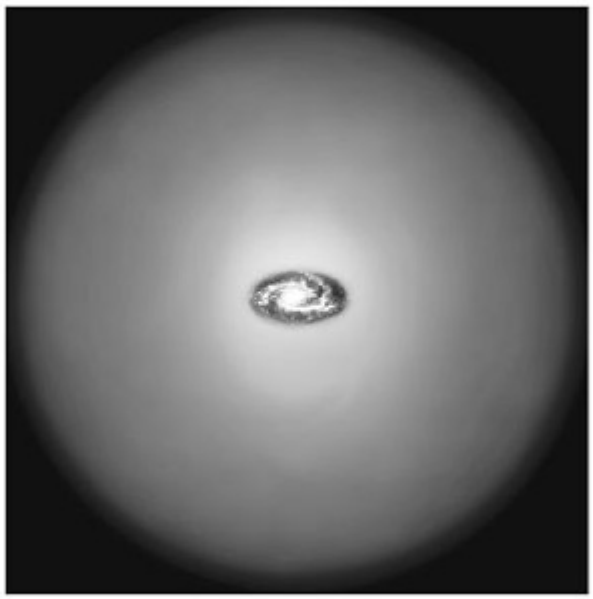


Figure 10.5 A galaxy immersed in a ball of dark matter (with the dark matter artificially highlighted to make it visible in the figure).